Practical No.: 5

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Problem Statement: Represent a graph of your college campus using an adjacency list /adjacency matrix.

Nodes should represent the various departments/institutes, and links should represent the distance between them.

Find the minimum spanning tree :

a. Using Kruskal's algorithm.

b. Using Prim's algorithm

Code:

```
INF = 999999

graph = [
  [0, 4, 2, 0, 0],
  [4, 0, 1, 5, 0],
  [2, 1, 0, 8, 10],
  [0, 5, 8, 0, 2],
  [0, 0, 10, 2, 0]
]

vertices = ["Computer", "IT", "Mechanical", "Library", "Office"]
V = len(vertices)

# ---------- Prim ---------- #

def prim():
  selected = [False] * V
  selected[0] = True
  edges = 0
```

```
    cost = 0

    print("\n Minimum Spanning Tree using Prim\n")

    while edges < V - 1:
        minimum = INF
        x = 0
        y = 0

        for i in range(V):
            if selected[i]:
                for j in range(V):
                    if (not selected[j]) and graph[i][j]:
                        if graph[i][j] < minimum:
                            minimum = graph[i][j]
                            x = i
                            y = j

        print(vertices[x], "--", vertices[y], "=", graph[x][y])

        cost += graph[x][y]
        selected[y] = True
        edges += 1

    print("\nMinimum Cost =", cost)

# ---------- Kruskal ---------- #

parent = []

def find(i):
```

```
    while parent[i] != i:
        i = parent[i]
    return i

def union(i, j):
    a = find(i)
    b = find(j)
    parent[a] = b

def kruskal():
    edges = []

    for i in range(V):
        for j in range(i + 1, V):
            if graph[i][j] != 0:
                edges.append((graph[i][j], i, j))

    edges.sort()

    for i in range(V):
        parent.append(i)

    cost = 0
    count = 0

    print("\nMinimum Spanning Tree using Kruskal\n")

    for w, u, v in edges:
        if find(u) != find(v):
            union(u, v)
```

```
            print(vertices[u], "--", vertices[v], "=", w)

            cost += w
            count += 1

            if count == V - 1:
                break

    print("\nMinimum Cost =", cost)

# ---------- Main ---------- #

while True:
    print("\n1. Prim")
    print("2. Kruskal")
    print("3. Exit")

    ch = int(input("Enter Choice : "))

    if ch == 1:
        prim()

    elif ch == 2:
        parent.clear()
        kruskal()

    elif ch == 3:
        break

    else:
        print("Invalid Choice")
```

Output:

1. Prim

2. Kruskal

3. Exit

Enter Choice: 1

Minimum Spanning Tree using Prim

Computer -- Mechanical = 2

Mechanical -- IT = 1

IT -- Library = 5

Library -- Office = 2

Minimum Cost = 10

1. Prim

2. Kruskal

3. Exit

Enter Choice: 2

Minimum Spanning Tree using Kruskal

IT -- Mechanical = 1

Computer -- Mechanical = 2

Library -- Office = 2

IT -- Library = 5

Minimum Cost = 10

1. Prim

2. Kruskal

3. Exit

Enter Choice : 3